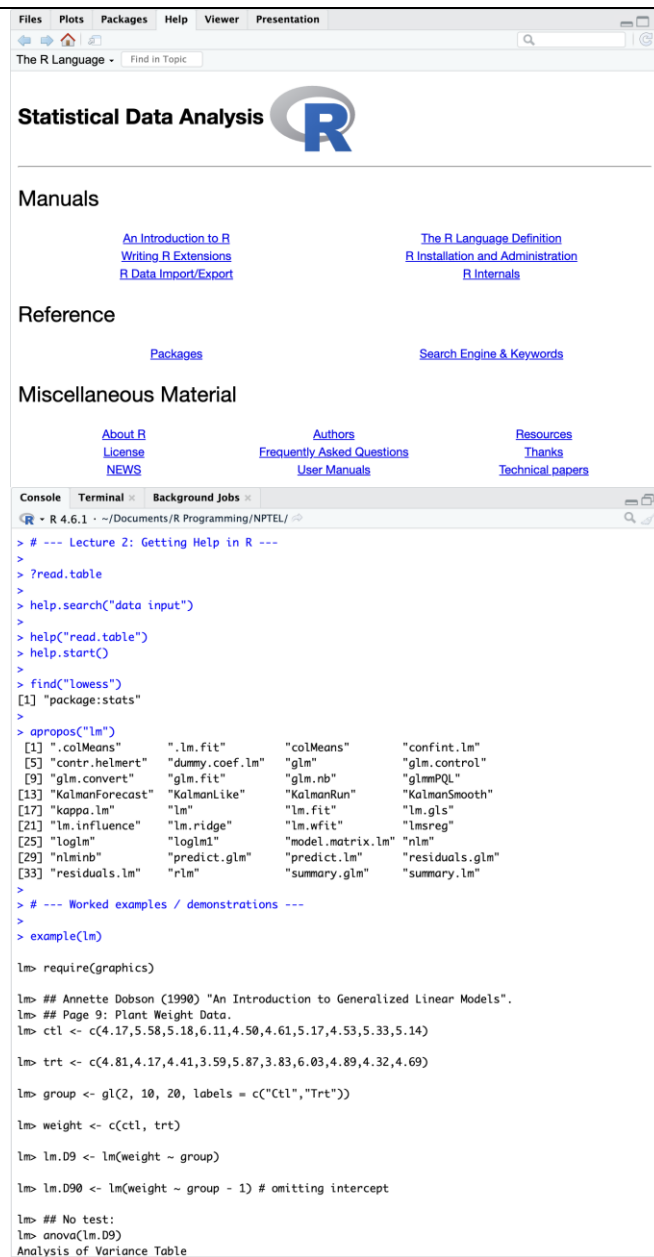


Name :	Siddhanth Kaushik Mungekar
Roll Number :	23102B0030
Branch :	CMPN/B

Code	 The screenshot shows the R Statistical Data Analysis website interface. The top navigation bar includes 'Files', 'Plots', 'Packages', 'Help', 'Viewer', and 'Presentation'. Below the navigation bar is a search bar and the title 'Statistical Data Analysis' with the R logo. The main content area is divided into three sections: 'Manuals', 'Reference', and 'Miscellaneous Material'. The 'Manuals' section contains links to 'An Introduction to R', 'Writing R Extensions', 'R Data Import/Export', 'The R Language Definition', 'R Installation and Administration', and 'R Internals'. The 'Reference' section contains links to 'Packages' and 'Search Engine & Keywords'. The 'Miscellaneous Material' section contains links to 'About R', 'License', 'NEWS', 'Authors', 'Frequently Asked Questions', 'User Manuals', 'Resources', 'Thanks', and 'Technical papers'. Below the website content is a console window showing R code and output. The code includes comments for 'Lecture 2: Getting Help in R', a search for 'data input', a search for 'lowess', and a list of R packages. The output shows the results of the search and the list of packages. <pre> R - R 4.6.1 - ~/Documents/R Programming/NPTEL/ > # --- Lecture 2: Getting Help in R --- > > ?read.table > > help.search("data input") > > help("read.table") > help.start() > > find("lowess") [1] "package:stats" > > apropos("lm") [1] ".lm.fit" "colMeans" "confint.lm" [5] "contr.helmert" "dummy.coef.lm" "glm" "glm.control" [9] "glm.convert" "glm.fit" "glm.nb" "glmPQL" [13] "KalmanForecast" "KalmanLike" "KalmanRun" "KalmanSmooth" [17] "kappa.lm" "lm" "lm.fit" "lm.gls" [21] "lm.influence" "lm.ridge" "lm.wfit" "lmsreg" [25] "loglm" "loglm1" "model.matrix.lm" "nlm" [29] "nlminb" "predict.glm" "predict.lm" "residuals.glm" [33] "residuals.lm" "rlm" "summary.glm" "summary.lm" > > # --- Worked examples / demonstrations --- > > example(lm) lm> require(graphics) lm> ## Annette Dobson (1990) "An Introduction to Generalized Linear Models". lm> ## Page 9: Plant Weight Data. lm> ctl <- c(4.17,5.58,5.18,6.11,4.50,4.61,5.17,4.53,5.33,5.14) lm> trt <- c(4.81,4.17,4.41,3.59,5.87,3.83,6.03,4.89,4.32,4.69) lm> group <- gl(2, 10, 20, labels = c("Ctl","Trt")) lm> weight <- c(ctl, trt) lm> lm.D9 <- lm(weight ~ group) lm> lm.D90 <- lm(weight ~ group - 1) # omitting intercept lm> ## No test: lm> anova(lm.D9) Analysis of Variance Table </pre>
------	--

```

Console Terminal Background Jobs
R 4.6.1 - ~/Documents/R Programming/NPTEL/
Analysis of Variance Table

Response: weight
      Df Sum Sq Mean Sq F value Pr(>F)
group    1  0.6882  0.68820   1.4191  0.249
Residuals 18  8.7293  0.48496

lm> summary(lm.D90)

Call:
lm(formula = weight ~ group - 1)

Residuals:
    Min       1Q   Median       3Q      Max
-1.0710 -0.4938  0.0685  0.2462  1.3690

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
groupCt1    5.0320      0.2202   22.85 9.55e-15 ***
groupTrt    4.6610      0.2202   21.16 3.62e-14 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.6964 on 18 degrees of freedom
Multiple R-squared:  0.9818,    Adjusted R-squared:  0.9798
F-statistic: 485.1 on 2 and 18 DF,  p-value: < 2.2e-16

lm> ## End(No test)
lm> opar <- par(mfrow = c(2,2), oma = c(0, 0, 1.1, 0))

lm> plot(lm.D9, las = 1) # Residuals, Fitted, ...
Hit <Return> to see next plot:

lm> par(opar)

lm> ## Don't show:
lm> ## model frame :
lm> stopifnot(identical(lm(weight ~ group, method = "model.frame"),
lm+               model.frame(lm.D9)))

lm> ## End(Don't show)
lm> ### less simple examples in "See Also" above
lm>
lm>
lm>

```

```

Console Terminal Background Jobs
R 4.6.1 - ~/Documents/R Programming/NPTEL/
> demo(persp)

demo(persp)
-----

Type <Return> to start :

> ### Demos for persp() plots -- things not in example(persp)
> ### -----
>
> require(datasets)
> require(grDevices); require(graphics)
> ## (1) The Obligatory Mathematical surface.
> ## Rotated sinc function.
>
> x <- seq(-10, 10, length.out = 50)
>
> y <- x
>
> rotsinc <- function(x,y)
+ {
+   sinc <- function(x) { y <- sin(x)/x ; y[is.na(y)] <- 1; y }
+   10 * sinc( sqrt(x^2+y^2) )
+ }
> sinc.exp <- expression(z == Sinc(sqrt(x^2 + y^2)))
> z <- outer(x, y, rotsinc)
> oldpar <- par(bg = "white")
> persp(x, y, z, theta = 30, phi = 30, expand = 0.5, col = "lightblue")
Hit <Return> to see next plot: demo(graphics)
> title(sub=".")## work around persp+plotmath bug
> title(main = sinc.exp)
> persp(x, y, z, theta = 30, phi = 30, expand = 0.5, col = "lightblue",
+       ltheta = 120, shade = 0.75, ticktype = "detailed",
+       xlab = "X", ylab = "Y", zlab = "Z")
Hit <Return> to see next plot:

```

ConsoleTerminalBackground Jobs

R 4.6.1 - ~/Documents/R Programming/NPTEL/Hit <Return> to see next plot:

> title(sub=".")## work around persp-plotmath bug

> title(main = sinc.exp)

> ## (2) Visualizing a simple DEM model
>
> z <- 2 * volcano # Exaggerate the relief

> x <- 10 * (1:nrow(z)) # 10 meter spacing (S to N)

> y <- 10 * (1:ncol(z)) # 10 meter spacing (E to W)

> persp(x, y, z, theta = 120, phi = 15, scale = FALSE, axes = FALSE)
Hit <Return> to see next plot:

> ## (3) Now something more complex
> ## We border the surface, to make it more "slice like"
> ## and color the top and sides of the surface differently.
>
> z0 <- min(z) - 20

> z <- rbind(z0, cbind(z0, z, z0), z0)

> x <- c(min(x) - 1e-10, x, max(x) + 1e-10)

> y <- c(min(y) - 1e-10, y, max(y) + 1e-10)

> fill <- matrix("green3", nrow = nrow(z)-1, ncol = ncol(z)-1)

> fill[, i2 <- c(1,ncol(fill))] <- "gray"

> fill[,i1 <- c(1,nrow(fill)),] <- "gray"

> par(bg = "lightblue")

> persp(x, y, z, theta = 120, phi = 15, col = fill, scale = FALSE, axes = FALSE)
Hit <Return> to see next plot: # --- Lecture 3: Packages and Libraries ---

> title(main = "Maunga Whau\nOne of 50 Volcanoes in the Auckland Region.",
+ font.main = 4)

> par(bg = "slategray")

> persp(x, y, z, theta = 135, phi = 30, col = fill, scale = FALSE,
+ ltheta = -120, lphi = 15, shade = 0.65, axes = FALSE)
Hit <Return> to see next plot:

> ## Don't draw the grid lines : border = NA
> persp(x, y, z, theta = 135, phi = 30, col = "green3", scale = FALSE,
+ ltheta = -120, lphi = 15, shade = 0.75, border = NA, box = FALSE)
Hit <Return> to see next plot: library(spatial)

> ## 'color gradient in the soil' :
> fcol <- fill ; fcol[] <- terrain.colors(nrow(fcol))

> persp(x, y, z, theta = 135, phi = 30, col = fcol, scale = FALSE,
+ ltheta = -120, shade = 0.3, border = NA, box = FALSE)
Hit <Return> to see next plot:

> ## 'image like' colors on top :
> fcol <- fill

> zi <- volcano[-1,-1] + volcano[-1,-61] +
+ volcano[-87,-1] + volcano[-87,-61] ## / 4

> fcol[-i1,-i2] <-
+ terrain.colors(20)[cut(zi,
+ stats::quantile(zi, seq(0,1, length.out = 21)),
+ include.lowest = TRUE)]

> persp(x, y, 2*z, theta = 110, phi = 40, col = fcol, scale = FALSE,
+ ltheta = -120, shade = 0.4, border = NA, box = FALSE)
Hit <Return> to see next plot: library(C)

> ## reset par():
> par(oldpar)
>
> packageDescription("spatial")
Package: spatial
Priority: recommended
Version: 7.3-19
Date: 2026-08-01
Depends: R (>= 3.0.0), graphics, stats, utils
Authors@R: c(person("Brian", "Ripley", role = c("aut", "cre", "cph"), email =
"Brian.Ripley@R-project.org"), person("Roger", "Bivand", role = "ctb"),
person("William", "Venables", role = "cph"))
Description: Functions for kriging and point pattern analysis.
Title: Functions for Kriging and Point Pattern Analysis
LazyData: yes

ConsoleTerminalBackground Jobs

R 4.6.1 - ~/Documents/R Programming/NPTEL/Hit <Return> to see next plot:

Authors@R: c(person("Brian", "Ripley", role = c("aut", "cre", "cph"), email =
"Brian.Ripley@R-project.org"), person("Roger", "Bivand", role = "ctb"),
person("William", "Venables", role = "cph"))
Description: Functions for kriging and point pattern analysis.
Title: Functions for Kriging and Point Pattern Analysis
LazyData: yes
ByteCompile: yes
License: GPL-2 | GPL-3
URL: http://www.stats.ox.ac.uk/pub/MASS4/
NeedsCompilation: yes
Packaged: 2026-08-03 06:32:27 UTC; ripley
Author: Brian Ripley [aut, cre, cph], Roger Bivand [ctb], William Venables [cph]
Maintainer: Brian Ripley <Brian.Ripley@R-project.org>
Repository: CRAN
Date/Publication: 2026-08-03 09:45:17 UTC
Built: R 4.6.1; aarch64-apple-darwin23; 2026-08-03 13:53:56 UTC; unix
Archs: spatial.so.dSYM

-- File: /Library/Frameworks/R.framework/Versions/4.6/Resources/Library/spatial/Meta/package.rds

>
> library(help = spatial)
>
> # Install packages
> install.packages("boot")
trying URL 'https://cran.rstudio.com/bin/macosx/sonoma-arm64/contrib/4.6/boot_1.3-32.tgz'
Content type 'application/x-gzip' length 645963 bytes (630 KB)
downloaded 630 KB

The downloaded binary packages are in
/var/folders/g5/wdrg4s3dj3q4mjrpdhcg6vvr0000gn/T//RtmpI1UeJb/downloaded_packages
> install.packages("cluster")
trying URL 'https://cran.rstudio.com/bin/macosx/sonoma-arm64/contrib/4.6/cluster_2.1.8.3.tgz'
Content type 'application/x-gzip' length 634529 bytes (619 KB)
downloaded 619 KB

The downloaded binary packages are in
/var/folders/g5/wdrg4s3dj3q4mjrpdhcg6vvr0000gn/T//RtmpI1UeJb/downloaded_packages

> # View installed packages
> installed.packages()
Package
"abind"

Console	Terminal	Background Jobs
R 4.6.1 · ~/Documents/R Programming/NPTEL/		
<pre>> > # View installed packages > installed.packages()</pre>		
	Package	
abind	"abind"	
askpass	"askpass"	
backports	"backports"	
base	"base"	
base64enc	"base64enc"	
BiocGenerics	"BiocGenerics"	
BiocManager	"BiocManager"	
BiocVersion	"BiocVersion"	
bit	"bit"	
bit64	"bit64"	
bitops	"bitops"	
blob	"blob"	
boot	"boot"	
broom	"broom"	
bslib	"bslib"	
cachem	"cachem"	
callr	"callr"	
car	"car"	
carData	"carData"	
cellranger	"cellranger"	
checkmate	"checkmate"	
class	"class"	
cli	"cli"	
clipr	"clipr"	
cluster	"cluster"	
codetools	"codetools"	
colorspace	"colorspace"	
commonmark	"commonmark"	
compiler	"compiler"	
config	"config"	
conflicted	"conflicted"	
cowplot	"cowplot"	
cpp11	"cpp11"	
crayon	"crayon"	
curl	"curl"	
data.table	"data.table"	
datasets	"datasets"	
DBI	"DBI"	
dbplyr	"dbplyr"	
Deriv	"Deriv"	
digest	"digest"	
doby	"doby"	
dotty	"dotty"	
dplyr	"dplyr"	
EBImage	"EBImage"	
effsize	"effsize"	
ellipsis	"ellipsis"	
evaluate	"evaluate"	
farver	"farver"	
fastmap	"fastmap"	
fftwtools	"fftwtools"	
fontawesome	"fontawesome"	
forcats	"forcats"	
forecast	"forecast"	
foreign	"foreign"	
Formula	"Formula"	
fractdiff	"fractdiff"	
LibPath		
abind	"/Library/Frameworks/R.framework/Versions/4.6/Resources/Library"	
askpass	"/Library/Frameworks/R.framework/Versions/4.6/Resources/Library"	
backports	"/Library/Frameworks/R.framework/Versions/4.6/Resources/Library"	
base	"/Library/Frameworks/R.framework/Versions/4.6/Resources/Library"	
base64enc	"/Library/Frameworks/R.framework/Versions/4.6/Resources/Library"	
BiocGenerics	"/Library/Frameworks/R.framework/Versions/4.6/Resources/Library"	
BiocManager	"/Library/Frameworks/R.framework/Versions/4.6/Resources/Library"	
BiocVersion	"/Library/Frameworks/R.framework/Versions/4.6/Resources/Library"	
bit	"/Library/Frameworks/R.framework/Versions/4.6/Resources/Library"	
bit64	"/Library/Frameworks/R.framework/Versions/4.6/Resources/Library"	
bitops	"/Library/Frameworks/R.framework/Versions/4.6/Resources/Library"	
blob	"/Library/Frameworks/R.framework/Versions/4.6/Resources/Library"	
boot	"/Library/Frameworks/R.framework/Versions/4.6/Resources/Library"	
broom	"/Library/Frameworks/R.framework/Versions/4.6/Resources/Library"	
bslib	"/Library/Frameworks/R.framework/Versions/4.6/Resources/Library"	
cachem	"/Library/Frameworks/R.framework/Versions/4.6/Resources/Library"	
callr	"/Library/Frameworks/R.framework/Versions/4.6/Resources/Library"	
car	"/Library/Frameworks/R.framework/Versions/4.6/Resources/Library"	
carData	"/Library/Frameworks/R.framework/Versions/4.6/Resources/Library"	
cellranger	"/Library/Frameworks/R.framework/Versions/4.6/Resources/Library"	
checkmate	"/Library/Frameworks/R.framework/Versions/4.6/Resources/Library"	
class	"/Library/Frameworks/R.framework/Versions/4.6/Resources/Library"	
cli	"/Library/Frameworks/R.framework/Versions/4.6/Resources/Library"	
clipr	"/Library/Frameworks/R.framework/Versions/4.6/Resources/Library"	
cluster	"/Library/Frameworks/R.framework/Versions/4.6/Resources/Library"	
codetools	"/Library/Frameworks/R.framework/Versions/4.6/Resources/Library"	
colorspace	"/Library/Frameworks/R.framework/Versions/4.6/Resources/Library"	
commonmark	"/Library/Frameworks/R.framework/Versions/4.6/Resources/Library"	
compiler	"/Library/Frameworks/R.framework/Versions/4.6/Resources/Library"	

ConsoleTerminalBackground Jobs

R 4.6.1 - ~/Documents/R Programming/NPTEL/

config"/Library/Frameworks/R.framework/Versions/4.6/Resources/Library"
conflicted"/Library/Frameworks/R.framework/Versions/4.6/Resources/Library"
complot"/Library/Frameworks/R.framework/Versions/4.6/Resources/Library"
cppl1"/Library/Frameworks/R.framework/Versions/4.6/Resources/Library"
croyon"/Library/Frameworks/R.framework/Versions/4.6/Resources/Library"
curl"/Library/Frameworks/R.framework/Versions/4.6/Resources/Library"
data.table"/Library/Frameworks/R.framework/Versions/4.6/Resources/Library"
datasets"/Library/Frameworks/R.framework/Versions/4.6/Resources/Library"
DBI"/Library/Frameworks/R.framework/Versions/4.6/Resources/Library"
dplyr"/Library/Frameworks/R.framework/Versions/4.6/Resources/Library"
Deriv"/Library/Frameworks/R.framework/Versions/4.6/Resources/Library"
digest"/Library/Frameworks/R.framework/Versions/4.6/Resources/Library"
doby"/Library/Frameworks/R.framework/Versions/4.6/Resources/Library"
dotly"/Library/Frameworks/R.framework/Versions/4.6/Resources/Library"
dplyr"/Library/Frameworks/R.framework/Versions/4.6/Resources/Library"
dplyr"/Library/Frameworks/R.framework/Versions/4.6/Resources/Library"
EBImage"/Library/Frameworks/R.framework/Versions/4.6/Resources/Library"
effsize"/Library/Frameworks/R.framework/Versions/4.6/Resources/Library"
ellipsis"/Library/Frameworks/R.framework/Versions/4.6/Resources/Library"
evaluate"/Library/Frameworks/R.framework/Versions/4.6/Resources/Library"
farver"/Library/Frameworks/R.framework/Versions/4.6/Resources/Library"
fastmap"/Library/Frameworks/R.framework/Versions/4.6/Resources/Library"
fftwtools"/Library/Frameworks/R.framework/Versions/4.6/Resources/Library"
fontawesome"/Library/Frameworks/R.framework/Versions/4.6/Resources/Library"
forcats"/Library/Frameworks/R.framework/Versions/4.6/Resources/Library"
forecast"/Library/Frameworks/R.framework/Versions/4.6/Resources/Library"
foreign"/Library/Frameworks/R.framework/Versions/4.6/Resources/Library"
Formula"/Library/Frameworks/R.framework/Versions/4.6/Resources/Library"
fractiff"/Library/Frameworks/R.framework/Versions/4.6/Resources/Library"

VersionPriority
abind"1.4-8"NA
askpass"1.2.1"NA
backports"1.5.1"NA
base"4.6.1"base"
base64enc"0.1-6"NA
BioGenerics"0.58.1"NA
BioManager"1.30.27"NA
BioVersion"3.23.1"NA
bit"4.6.0"NA
bit64"4.8.4"NA
bitops"1.1-0"NA
blob"1.3-8"NA
boot"1.3-32"recommended"
broom"1.0.13"NA
bslib"0.12.0"NA
cachem"1.1.0"NA

ConsoleTerminalBackground Jobs

R 4.6.1 - ~/Documents/R Programming/NPTEL/

callr"3.8.0"NA
car"3.1-5"NA
carData"3.0-6"NA
cellranger"1.1.0"NA
checkmate"2.3-4"NA
class"7.3-24"recommended"
cli"3.6.6"NA
clipr"0.8.1"NA
cluster"2.1.8.3"recommended"
codetools"0.2-20"recommended"
colorspace"2.1-3"NA
commonmark"2.0-0"NA
compiler"4.6.1"base"
config"0.3-2"NA
conflicted"1.2-0"NA
complot"1.2-0"NA
cppl1"0.5-5"NA
croyon"1.5-3"NA
curl"8.0-0"NA
data.table"1.18.6.1"NA
datasets"4.6.1"base"
DBI"1.3-0"NA
dplyr"2.6-0"NA
Deriv"4.3-5"NA
digest"0.6.39"NA
doby"4.7.2"NA
dotly"0.1-0"NA
dplyr"1.2.1"NA
dplyr"1.3.3"NA
EBImage"4.54-0"NA
effsize"0.8.1"NA
ellipsis"0.3.3"NA
evaluate"1.0-5"NA
farver"2.1.2"NA
fastmap"1.2-0"NA
fftwtools"0.9-11"NA
fontawesome"0.5-3"NA
forcats"1.0-1"NA
forecast"0.9-2"NA
foreign"0.8-91"recommended"
Formula"1.2-6"NA
fractiff"1.5-4"NA
Depends
abind"R (>= 1.5.0)"
askpass"NA
backports"R (>= 3.0.0)"

ConsoleTerminalBackground Jobs

R 4.6.1 - ~/Documents/R Programming/NPTEL/

baseNA
base64enc"R (>= 2.9.0)"
BioGenerics"R (>= 4.0.0), methods, utils, graphics, stats, generics"
BioManagerNA
BioVersion"R (>= 4.0.0)"
bit"R (>= 3.4.0)"
bit64"R (>= 3.5.0)"
bitopsNA
blob"R (>= 3.0.0), graphics, stats"
broom"R (>= 4.1)"
bslib"R (>= 2.10)"
cachemNA
callr"R (>= 3.4)"
car"R (>= 3.5.0), carData (>= 3.0.0)"
carData"R (>= 3.5.0)"
cellranger"R (>= 3.0.0)"
checkmate"R (>= 3.0.0)"
class"R (>= 3.0.0), stats, utils"
cli"R (>= 3.4)"
clipr"R (>= 3.5.0)"
cluster"R (>= 2.1)"
codetools"R (>= 3.0.0), methods"
colorspaceNA
commonmarkNA
compilerNA
configNA
conflicted"R (>= 3.2)"
complot"R (>= 3.5.0)"
cppl1"R (>= 4.0.0)"
croyonNA
curl"R (>= 3.0.0)"
data.table"R (>= 3.4.0)"
datasetsNA
DBI"methods, R (>= 3.0.0)"
dplyr"R (>= 4.1)"
Deriv"Rcpp, R (>= 4.5)"
digest"R (>= 3.3.0)"
doby"R (>= 4.2.0), methods"
dotlyNA
dplyr"R (>= 4.1.0)"
dplyr"R (>= 4.0)"
EBImage"methods"
effsize"R (>= 3.2)"
ellipsis"R (>= 3.6.0)"
evaluate

ConsoleTerminalBackground Jobs

R 4.6.1 - ~/Documents/R Programming/NPTEL/

farverNA
fastmapNA
fftwtools"R (>= 3.0)"
fontawesome"R (>= 3.3.0)"
forcats"R (>= 4.1)"
forecast"R (>= 4.1.0)"
foreign"R (>= 4.0.0)"
Formula"R (>= 2.0.0), stats"
fractiffNA
Imports
abind"methods, utils"
askpass"sys (>= 2.1)"
backportsNA
baseNA
base64encNA
BioGenerics"methods, utils, graphics, stats"
BioManager"utils"
BioVersionNA
bitNA
bit64"bit (>= 4.0.0), graphics, methods, stats, utils"
bitopsNA
blob"methods, rlang, vctrs (>= 0.2.1)"
bootNA
broom"backports, cli, dplyr (>= 1.0.0), generics (>= 0.0.2), glue, lifecycle, purrr, rlang (>= 1.1.0), stringr, tibble (>= 3.0.0), vctrs (>= 1.0.0)"
bslib"base64enc, cachem, fastmap (>= 1.1.1), grDevices, htmltools (>= 0.5.8), jquerylib (>= 0.1.3), jsonlite, lifecycle, memoise (>= 2.0.1), mime, rlang, sass (>= 0.4.9)"
cachem"rlang, fastmap (>= 1.2.0)"
callr"otel (>= 0.2.0), processx (>= 3.6.1), R6, utils"
car"abind, Formula, MASS, mgcv, mnet, pbkrtest (>= 0.4-4), nqantreg, grDevices, util
s, stats, graphics, lme4 (>= v1.1-27.1), nlme, scales"
carDataNA
cellranger"rematch, tibble"
checkmate"backports (>= 1.1.0), utils"
classNA
cli"utils"
clipr"utils"
cluster"graphics, grDevices, stats, utils"
codetoolsNA
colorspace"graphics, grDevices, stats"
commonmarkNA
compilerNA
config"yaml (>= 2.1.19)"
conflicted"cli (>= 3.4.0), memoise, rlang (>= 1.0.0)"
complot"ggplot2 (>= 3.5.2), grid, gtable, grDevices, methods, rlang, nscales"
cppl1NA

The image displays a RStudio window with the title "R 4.6.1 - ~/Documents/R Programming/NPTEL/". The console pane on the left shows the output of the `install.packages()` function, listing 100 installed packages. The packages are listed in a single column, with their names and version numbers. The list includes: crayon, curl, data.table, datasets, DBI, dplyr, lifecycle, magrittr, methods, pillar, R6, rlang, tidyr, tidyselect, utils, vctrs, and many others. The console window is titled "Console" and shows the output of the `install.packages()` function. The RStudio window title is "R 4.6.1 - ~/Documents/R Programming/NPTEL/".

Console	Terminal	Background Jobs
R 4.6.1 - ~/Documents/R Programming/NPTEL/ ↗		
bitops	NA	
blob	NA	
boot	NA	
broom	NA	
bslib	NA	
cachem	NA	
callr	NA	
car	NA	
carData	NA	
cellranger	NA	
checkmate	NA	
class	NA	
cli	NA	
clipr	NA	
cluster	"mvoutlier, fpc, ellipse, sfsmisc"	
codetools	NA	
colorspace	NA	
commonmark	NA	
compiler	NA	
config	NA	
conflicted	NA	
cowplot	NA	
cpp11	NA	
crayon	NA	
curl	NA	
data.table	NA	
datasets	NA	
DBI	NA	
dbplyr	NA	
Deriv	NA	
digest	NA	
doBy	NA	
dotty	NA	
dplyr	NA	
dtplyr	NA	
EBImage	NA	
effsize	NA	
ellipsis	NA	
evaluate	NA	
farver	NA	
fastmap	NA	
fftwtools	NA	
fontawesome	NA	
forcats	NA	
forecast	NA	
foreign	NA	

Console	Terminal	Background Jobs
R 4.6.1 - ~/Documents/R Programming/NPTEL/ ↗		
Formula	NA	
fractdiff	NA	
	License	License_is_FOSS
abind	"MIT + file LICENSE"	NA
askpass	"MIT + file LICENSE"	NA
backports	"GPL-2 GPL-3"	NA
base	"Part of R 4.6.1"	NA
base64enc	"GPL-2 GPL-3"	NA
BiocGenerics	"Artistic-2.0"	NA
BiocManager	"Artistic-2.0"	NA
BiocVersion	"Artistic-2.0"	NA
bit	"GPL-2 GPL-3"	NA
bit64	"GPL-2 GPL-3"	NA
bitops	"GPL (>= 2)"	NA
blob	"MIT + file LICENSE"	NA
boot	"Unlimited"	NA
broom	"MIT + file LICENSE"	NA
bslib	"MIT + file LICENSE"	NA
cachem	"MIT + file LICENSE"	NA
callr	"MIT + file LICENSE"	NA
car	"GPL (>= 2)"	NA
carData	"GPL (>= 2)"	NA
cellranger	"MIT + file LICENSE"	NA
checkmate	"BSD_3_clause + file LICENSE"	NA
class	"GPL-2 GPL-3"	NA
cli	"MIT + file LICENSE"	NA
clipr	"GPL-3"	NA
cluster	"GPL (>= 2)"	NA
codetools	"GPL"	NA
colorspace	"BSD_3_clause + file LICENSE"	NA
commonmark	"BSD_2_clause + file LICENSE"	NA
compiler	"Part of R 4.6.1"	NA
config	"GPL-3"	NA
conflicted	"MIT + file LICENSE"	NA
cowplot	"GPL-2"	NA
cpp11	"MIT + file LICENSE"	NA
crayon	"MIT + file LICENSE"	NA
curl	"MIT + file LICENSE"	NA
data.table	"MPL-2.0 file LICENSE"	NA
datasets	"Part of R 4.6.1"	NA
DBI	"LGPL (>= 2.1)"	NA
dbplyr	"MIT + file LICENSE"	NA
Deriv	"GPL (>= 3)"	NA
digest	"GPL (>= 2)"	NA
doBy	"GPL (>= 2)"	NA
dotty	"MIT + file LICENSE"	NA

Console	Terminal	Background Jobs
R 4.6.1 - ~/Documents/R Programming/NPTEL/ ↗		
dplyr	"MIT + file LICENSE"	NA
dtplyr	"MIT + file LICENSE"	NA
EImage	"LGPL"	NA
effsize	"GPL-2"	NA
ellipsis	"MIT + file LICENSE"	NA
evaluate	"MIT + file LICENSE"	NA
farver	"MIT + file LICENSE"	NA
fastmap	"MIT + file LICENSE"	NA
fftwtools	"GPL (>= 2)"	NA
fontawesome	"MIT + file LICENSE"	NA
forcats	"MIT + file LICENSE"	NA
forecast	"GPL-3"	NA
foreign	"GPL (>= 2)"	NA
Formula	"GPL-2 GPL-3"	NA
fracdiff	"GPL (>= 2)"	NA
abind	License_restricts_use OS.type MD5sum NeedsCompilation	
askpass	NA	NA "no"
backports	NA	NA NA "yes"
base	NA	NA NA NA
base64enc	NA	NA NA "yes"
BioGenerics	NA	NA NA "no"
BiocManager	NA	NA NA "no"
BiocVersion	NA	NA NA "no"
bit	NA	NA NA "yes"
bit64	NA	NA NA "yes"
bitops	NA	NA NA "yes"
blob	NA	NA NA "no"
boot	NA	NA NA "no"
broom	NA	NA NA "no"
bslib	NA	NA NA "no"
cachem	NA	NA NA "yes"
callr	NA	NA NA "no"
car	NA	NA NA "no"
carData	NA	NA NA "no"
cellranger	NA	NA NA "no"
checkmate	NA	NA NA "yes"
class	NA	NA NA "yes"
cli	NA	NA NA "yes"
clipr	NA	NA NA "no"
cluster	NA	NA NA "yes"
codetools	NA	NA NA "no"
colorspace	NA	NA NA "yes"
commonmark	NA	NA NA "yes"
compiler	NA	NA NA NA
config	NA	NA NA "no"
Console	Terminal	Background Jobs
R 4.6.1 - ~/Documents/R Programming/NPTEL/ ↗		
conflicted	NA	NA NA "no"
complot	NA	NA NA "no"
cpp11	NA	NA NA "no"
croyon	NA	NA NA "no"
curl	NA	NA NA "yes"
data.table	NA	NA NA "yes"
datasets	NA	NA NA NA
DBI	NA	NA NA "no"
dbplyr	NA	NA NA "no"
Deriv	NA	NA NA "yes"
digest	NA	NA NA "yes"
doBy	NA	NA NA "no"
dotly	NA	NA NA "no"
dplyr	NA	NA NA "yes"
dtplyr	NA	NA NA "no"
EImage	NA	NA NA "yes"
effsize	NA	NA NA "no"
ellipsis	NA	NA NA "no"
evaluate	NA	NA NA "no"
farver	NA	NA NA "yes"
fastmap	NA	NA NA "yes"
fftwtools	NA	NA NA "yes"
fontawesome	NA	NA NA "no"
forcats	NA	NA NA "no"
forecast	NA	NA NA "yes"
foreign	NA	NA NA "yes"
Formula	NA	NA NA "no"
fracdiff	NA	NA NA "yes"
abind	Built	Published
askpass	"R 4.6.0; ; 2026-04-25 15:40:28 UTC; unix"	NA
backports	"R 4.6.0; aarch64-apple-darwin23; 2026-04-25 23:17:34 UTC; unix"	NA
base	"R 4.6.1; ; 2026-06-25 21:53:52 UTC; unix"	NA
base64enc	"R 4.6.0; aarch64-apple-darwin23; 2026-04-25 15:38:42 UTC; unix"	NA
BioGenerics	"R 4.6.0; ; 2026-05-14 10:19:55 UTC; unix"	NA
BiocManager	"R 4.6.0; ; 2026-04-25 15:56:04 UTC; unix"	NA
BiocVersion	"R 4.6.0; ; 2026-03-01 08:32:25 UTC; unix"	NA
bit	"R 4.6.0; aarch64-apple-darwin23; 2026-04-25 15:45:18 UTC; unix"	NA
bit64	"R 4.6.1; aarch64-apple-darwin23; 2026-08-20 08:52:04 UTC; unix"	NA
bitops	"R 4.6.1; aarch64-apple-darwin23; 2026-07-30 18:09:30 UTC; unix"	NA
blob	"R 4.6.0; ; 2026-04-25 23:18:16 UTC; unix"	NA
boot	"R 4.6.0; ; 2026-04-25 15:42:15 UTC; unix"	NA
broom	"R 4.6.0; ; 2026-05-14 05:30:47 UTC; unix"	NA
bslib	"R 4.6.1; ; 2026-08-04 04:11:06 UTC; unix"	NA
cachem	"R 4.6.0; aarch64-apple-darwin23; 2026-04-25 23:13:45 UTC; unix"	NA
callr	"R 4.6.0; ; 2026-06-05 15:30:17 UTC; unix"	NA

```
Console Terminal Background Jobs
R 4.6.1 ~/Documents/R Programming/NPTEL/ >
car "R 4.6.0; ; 2026-04-25 14:03:03 UTC; unix" NA
carData "R 4.6.0; ; 2026-04-25 15:41:28 UTC; unix" NA
cellranger "R 4.6.0; ; 2026-04-26 05:17:23 UTC; unix" NA
checkmate "R 4.6.0; aarch64-apple-darwin23; 2026-04-25 15:45:36 UTC; unix" NA
class "R 4.6.1; aarch64-apple-darwin23; 2026-08-03 13:53:56 UTC; unix" NA
cli "R 4.6.0; aarch64-apple-darwin23; 2026-04-25 15:39:10 UTC; unix" NA
clipr "R 4.6.0; ; 2026-05-25 08:30:02 UTC; unix" NA
cluster "R 4.6.1; aarch64-apple-darwin23; 2026-07-30 18:10:02 UTC; unix" NA
codetools "R 4.6.1; ; 2026-06-25 21:54:07 UTC; unix" NA
colorspace "R 4.6.1; aarch64-apple-darwin23; 2026-07-12 10:21:47 UTC; unix" NA
commonmark "R 4.6.0; aarch64-apple-darwin23; 2026-04-25 09:27:57 UTC; unix" NA
compiler "R 4.6.1; ; 2026-06-25 21:53:16 UTC; unix" NA
config "R 4.6.0; ; 2026-04-25 15:49:14 UTC; unix" NA
conflicted "R 4.6.0; ; 2026-04-26 02:30:17 UTC; unix" NA
complot "R 4.6.0; ; 2026-04-26 11:08:33 UTC; unix" NA
cpp11 "R 4.6.0; ; 2026-05-06 16:42:29 UTC; unix" NA
croyon "R 4.6.0; ; 2026-04-25 15:44:21 UTC; unix" NA
curl "R 4.6.1; aarch64-apple-darwin23; 2026-08-25 18:01:50 UTC; unix" NA
data.table "R 4.6.1; aarch64-apple-darwin23; 2026-08-24 13:00:59 UTC; unix" NA
datasets "R 4.6.1; ; 2026-06-25 21:53:43 UTC; unix" NA
DBI "R 4.6.0; ; 2026-04-25 15:47:33 UTC; unix" NA
dbplyr "R 4.6.0; ; 2026-06-18 00:56:28 UTC; unix" NA
Deriv "R 4.6.1; aarch64-apple-darwin23; 2026-09-03 18:24:15 UTC; unix" NA
digest "R 4.6.0; aarch64-apple-darwin23; 2026-04-25 15:38:59 UTC; unix" NA
doby "R 4.6.1; ; 2026-07-01 12:35:32 UTC; unix" NA
dotly "R 4.6.0; ; 2026-04-25 17:16:06 UTC; unix" NA
dplyr "R 4.6.0; aarch64-apple-darwin23; 2026-04-26 11:09:07 UTC; unix" NA
dtplyr "R 4.6.0; ; 2026-04-26 11:14:03 UTC; unix" NA
EBImage "R 4.6.0; aarch64-apple-darwin23; 2026-04-29 04:41:30 UTC; unix" NA
effsize "R 4.6.0; ; 2026-04-25 16:07:37 UTC; unix" NA
ellipsis "R 4.6.0; ; 2026-04-25 15:58:29 UTC; unix" NA
evaluate "R 4.6.0; ; 2026-04-25 15:39:47 UTC; unix" NA
farver "R 4.6.0; aarch64-apple-darwin23; 2026-04-25 15:40:10 UTC; unix" NA
fastmap "R 4.6.0; aarch64-apple-darwin23; 2026-04-25 15:38:44 UTC; unix" NA
fftwtools "R 4.6.0; aarch64-apple-darwin23; 2026-04-25 15:59:41 UTC; unix" NA
fontawesome "R 4.6.0; ; 2026-04-26 02:18:26 UTC; unix" NA
forcats "R 4.6.0; ; 2026-04-26 11:09:08 UTC; unix" NA
forecast "R 4.6.0; aarch64-apple-darwin23; 2026-04-26 11:10:38 UTC; unix" NA
foreign "R 4.6.1; aarch64-apple-darwin23; 2026-06-25 21:54:07 UTC; unix" NA
Formula "R 4.6.1; ; 2026-08-03 09:17:05 UTC; unix" NA
fractdiff "R 4.6.0; aarch64-apple-darwin23; 2026-04-28 06:38:56 UTC; unix" NA
[ reached 'max' / getOption("max.print") -- omitted 142 rows ]
>
> # Remove a package
> remove.packages("cluster")
Removing package from '/Library/Frameworks/R.framework/Versions/4.6/Resources/Library'
> # Update a package
> update.packages("cluster")

Warning message:
In .available.bboth(repos, method, ...) :
  Some listed binary packages have no source

>
> # Unload a package
> detach("package:cluster", unload = TRUE)

Error in detach("package:cluster", unload = TRUE) :
  invalid 'name' argument

> |
```